Three-Year Dementia Risk Prediction from Multimodal Clinical Data: Does Cardiovascular Information Add Value Beyond Cognition?

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by

Akula Akhil Vighnesh - CB.SC.U4AIE23303

Vada Gouri Hansika Reddy - CB.SC.U4AIE23304

Pilli Venkata Bhargav Reddy - CB.SC.U4AIE23329

Marella Venkata Mahidhar Reddy - CB.SC.U4AIE23354



DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE CAMPUS (INDIA)

October 2026

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE CAMPUS



BONAFIDE CERTIFICATE

This is to certify that the thesis entitled "**Three-Year Dementia Risk Prediction from Multimodal Clinical Data: Does Cardiovascular Information Add Value Beyond Cognition?**" submitted by **Akula Akhil Vighnesh, CB.SC.U4AIE23303**, **Vada Gouri Hansika Reddy, CB.SC.U4AIE23304**, **Pilli Venkata Bhargav Reddy, CB.SC.U4AIE23329** and **Marella Venkata Mahidhar Reddy, CB.SC.U4AIE23354** for the award of the **Degree of Bachelor of Technology** under the **Faculty of Engineering** is a bonafide record of the work carried out by them under my guidance and supervision at "Department of Artificial Intelligence", Amrita Vishwa Vidyapeetham, Coimbatore Campus.

Submitted for the examination held on [DATE].

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Dr. Manoj Bhat K
Thesis Advisor,
[Designation],
Department of Artificial Intelligence,
Amrita Vishwa Vidyapeetham,
Coimbatore Campus,
India.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE CAMPUS

DECLARATION

We, **Akula Akhil Vighnesh, CB.SC.U4AIE23303**, **Vada Gouri Hansika Reddy, CB.SC.U4AIE23304**, **Pilli Venkata Bhargav Reddy, CB.SC.U4AIE23329** and **Marella Venkata Mahidhar Reddy, CB.SC.U4AIE23354** hereby declare that this thesis entitled "**Three-Year Dementia Risk Prediction from Multimodal Clinical Data: Does Cardiovascular Information Add Value Beyond Cognition?**" is the record of the original work done by us under the guidance of **Dr. Manoj Bhat K**, [Designation], Department of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore Campus. To the best of our knowledge, this work has not formed the basis for the award of any degree/diploma/associateship/fellowship or a similar award to any candidate in any University.

Place: Coimbatore, India **Signature of the Students**
Date: [DATE]

COUNTERSIGNED

Dr. K.P Soman
Professor and Dean,
Department of Artificial Intelligence
Amrita Vishwa Vidyapeetham,
Coimbatore Campus,
India.

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[Insert here, word for word, the NACC acknowledgement statement and grant information required by the NACC data use agreement, as published on naccdata.org. Do not paraphrase it.]

AKULA AKHIL VIGHNESH
VADA GOURI HANSIKA REDDY
PILLI VENKATA BHARGAV REDDY
MARELLA VENKATA MAHIDHAR REDDY

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Abbreviations

APOE	–	Apolipoprotein E
AUPRC	–	Area Under the Precision–Recall Curve
AUROC	–	Area Under the Receiver Operating Characteristic Curve
BM25	–	Best Matching 25 (keyword ranking function)
BMI	–	Body Mass Index
CDR	–	Clinical Dementia Rating
CDR-SB	–	Clinical Dementia Rating, Sum of Boxes
CI	–	Confidence Interval
CNN	–	Convolutional Neural Network
DICOM	–	Digital Imaging and Communications in Medicine
FAQ	–	Functional Activities Questionnaire
FDG	–	Fluorodeoxyglucose
FT-Transformer	–	Feature Tokenizer Transformer
GDS	–	Geriatric Depression Scale
GRU	–	Gated Recurrent Unit
MCI	–	Mild Cognitive Impairment
MLP	–	Multilayer Perceptron
MMSE	–	Mini-Mental State Examination
MoCA	–	Montreal Cognitive Assessment
NACC	–	National Alzheimer's Coordinating Center
NPI-Q	–	Neuropsychiatric Inventory Questionnaire
PET	–	Positron Emission Tomography
RAG	–	Retrieval-Augmented Generation
SHAP	–	SHapley Additive exPlanations
SUVR	–	Standardised Uptake Value Ratio
UDS	–	Uniform Data Set
XGBoost	–	Extreme Gradient Boosting

List of symbols

$\mathbf{x}_{t_0}$	–	Feature vector at the index visit
t_0	–	Date of the index visit
y	–	Outcome: dementia diagnosis within three years (1) or not (0)
$\hat{p}$	–	Predicted probability of the outcome
s	–	Raw neuropsychological test score
$\hat{s}$	–	Expected test score of a cognitively normal person
σ_{res}	–	Residual standard deviation of the normative model
z	–	Demographically adjusted test score
β_k	–	Coefficients of the normative regression model
ϕ_j	–	SHAP contribution of feature j
ΔAUROC	–	Paired difference in AUROC between two models
p_{drop}	–	Probability of removing a category in modality dropout
B	–	Brier score

Abstract

Dementia develops over several years, and many kinds of information are routinely recorded about older adults who do not yet have it: demographic characteristics, cardiovascular and medical history, neuropsychological test results, a clinician's rating of daily functioning, the APOE genotype and, increasingly, amyloid and tau positron emission tomography (PET). It is not well established how much each of these sources contributes to short-term prediction once the others are known. This thesis develops and evaluates an explainable framework that estimates, for a person without dementia at a clinic visit, the probability of a dementia diagnosis within the following three years, and measures the contribution of each source, with particular attention to cardiovascular information.

The study uses the National Alzheimer's Coordinating Center database (57,038 participants, 217,598 visits). After harmonising neuropsychological tests from different form versions with demographically adjusted z-scores and constructing 48 features in five categories, 19,470 participants without dementia and with a known three-year outcome were analysed, of whom 2,818 were diagnosed with dementia. Participants were split by person, and nine whole centres were held out for evaluation. Five classifiers, from logistic regression to an FT-Transformer, achieved nearly identical discrimination (test AUROC 0.932 to 0.939; held-out centres 0.940 to 0.946). Within participants with mild cognitive impairment the AUROC was 0.84, and within cognitively normal participants 0.88, the latter based on few events. Most of the predictive signal came from current cognitive status; cardiovascular and medical history added about 0.001 AUROC beyond it, earlier visits added at most 0.005, and amyloid and tau PET measures did not produce a demonstrable gain. Training with modality dropout kept predictions calibrated when whole categories of input were missing. A separate PET image pipeline reproduced the PET core's amyloid measurements (correlation 0.98 over 318 scans). The models were integrated into a research prototype with plausibility checks and a retrieval-based evidence assistant, and the pipeline was audited by seventeen independent verification checks, which identified and corrected two sources of information leakage.

Keywords: dementia prediction, mild cognitive impairment, NACC, cardiovascular risk, explainable machine learning, SHAP, amyloid PET, retrieval-augmented generation.

Chapter 1

Introduction

1.1 Background and Motivation

Dementia is a clinical syndrome in which memory and other cognitive abilities decline to the point that a person can no longer manage daily life independently. Alzheimer's disease is its most common cause. The underlying brain changes begin years before the diagnosis, and many people pass through an intermediate stage, mild cognitive impairment (MCI), in which deficits are measurable but daily function is preserved. Not everyone with MCI progresses, and some people who appear cognitively normal decline within a few years. Estimating who is likely to progress in the near future is useful for monitoring, for planning care and for selecting participants for clinical trials.

A wide range of information is available for this purpose. Demographic characteristics, in particular age, are strong risk factors. Cardiovascular and metabolic conditions such as hypertension and diabetes have been associated with later dementia in epidemiological studies [1–4]. The APOE $\varepsilon 4$ allele is the major genetic risk factor for late-onset Alzheimer's disease [5]. Neuropsychological tests and clinical ratings measure the current state of cognition and function, and amyloid and tau PET imaging measures the pathology directly [6, 7]. What is less clear is how much each source adds once the others are known, and whether the added complexity of modern machine-learning models is justified for such data.

1.2 How the Research Question Developed

The project began with a broad aim: to predict cognitive decline from cardiovascular risk factors combined with longitudinal cognitive data, APOE and PET biomarkers, and to explain the predictions with supporting literature. Before any model was built, the data were audited in detail. The audit showed that the clinical diagnosis and the Clinical Dementia Rating are recorded consistently at every visit across twenty years of data, that roughly one in three participants with MCI progresses to dementia within three years,

and that most PET scans are too recent to observe long-term outcomes. On this basis the aim was narrowed to one precise and measurable question: the probability of a dementia diagnosis within three years of a clinic visit. The original cardiovascular hypothesis was retained, but as a question to be tested rather than an assumption: whether cardiovascular information adds predictive value once current cognition is known.

1.3 Changes from the Initial Proposal

The framework proposed at the zeroth review was designed before the data had been examined. Several of its components were changed once the data audit and the first experiments showed what the data could support. Table 1.1 records each change and its reason.

Table 1.1: Planned components at the zeroth review and what was implemented.

Planned	Implemented	Reason
Target: cognitive decline and progression, not yet fixed	Dementia diagnosis within three years; CDR-SB rise of one point built as a secondary label	Diagnosis is recorded consistently at every visit in every form version, and the three-year horizon balances events against loss to follow-up
Dataset: NACC via the ADSP-PHC harmonised release, NHANES	NACC Freeze 74 (UDS and SCAN PET tables, raw PET images)	The UDS release contains all required clinical variables with longitudinal visits and links to the PET data; NHANES does not provide longitudinal dementia diagnoses
XGBoost or CatBoost for tabular data	Logistic regression, random forest, XGBoost, MLP and FT-Transformer	A wider comparison answers whether model complexity helps; the models performed equally, so CatBoost was not added
LSTM or Transformer for cognitive trajectories	Trajectory features and a GRU on landmark visits	A GRU is the smaller recurrent model suited to short, irregular sequences of at most six visits; history added little
3D ResNet for PET with attention-based fusion	Small 3D CNN for amyloid status; late fusion of PET measures; modality dropout for missing data	Only 5 of 446 participants with images converted to dementia, so an image-based fusion model for progression could not be trained reliably; an early prototype of this kind gave unstable results (Section 1.3.1)
SHAP, Grad-CAM, attention weights, DiCE	SHAP and Grad-CAM	Counterfactual explanations suggest what to change in a patient, which the observational data cannot support
RAG grounding of predictions	Evidence assistant with hybrid retrieval and guardrails	Implemented as planned, with retrieval evaluated against a fixed question set
Target AUROC 0.70 to 0.78	AUROC 0.94 overall, 0.84 within MCI	The overall figure is higher than expected mainly because current cognitive status is a strong predictor; results are reported per diagnostic group for this reason

1.3.1 Preliminary Experiment

An early prototype followed the original plan closely: a 3D CNN on raw PET volumes fused with clinical features through cross-attention, predicting a rise of at least 0.5 in CDR-SB after one year. Only 198 participants had both a usable image and a one-year outcome, of whom 17 progressed. Under five-fold cross-validation the fused model reached a mean AUROC of 0.65 with fold values ranging from 0.40 to 0.86, and tabular models alone did no better than 0.78 with a standard deviation above 0.1. With so few events the estimates were too unstable to support any conclusion. This motivated the change to the full clinical cohort, a three-year dementia outcome, and a separate, validated treatment of the PET images.

1.4 Problem Statement

Let $\mathbf{x}_{t_0}$ denote the information available about a person without dementia at an index visit t_0: demographics, cardiovascular and medical history, neuropsychological test results, the clinician's rating, genetics and, where available, PET measures. The problem is to estimate

$$\hat{p} = P\left(y = 1 \mid \mathbf{x}_{t_0}\right), \tag{1.1}$$

where $y = 1$ if dementia is diagnosed within three years of t_0 and $y = 0$ if the person is followed for three years without such a diagnosis; to determine how much each category of information contributes to $\hat{p}$; and to do so without using any information that would not have been available at t_0.

1.5 Research Questions and Objectives

The study addresses six research questions:

RQ1. How accurately can a dementia diagnosis within three years be predicted for people without dementia, and how does this differ between cognitively normal people and people with MCI?

RQ2. Does cardiovascular and medical information improve the prediction beyond demographics, genetics and current cognition?

RQ3. Do amyloid and tau PET measures improve the prediction?

RQ4. Do deep-learning models outperform classical models on this kind of data?

RQ5. Does information from earlier visits improve the prediction beyond the current visit?

RQ6. Can a single model provide calibrated estimates when whole categories of information are missing?

The corresponding objectives were to construct a leakage-free cohort and outcome from the NACC database; to harmonise neuropsychological tests across form versions; to train and compare five classifiers under a person-level split with held-out centres; to measure the contribution of each category by ablation; to explain predictions with Shapley values; to test the added value of PET and of earlier visits; to validate the model in subgroups, for calibration and in sensitivity analyses; to build and validate a PET image pipeline; to integrate the model into a research prototype with an evidence assistant; and to verify the integrity of the entire pipeline with independent checks.

1.6 Contributions

- A reproducible pipeline that turns the NACC database into a leakage-free three-year prediction task, with test harmonisation across form versions.
- A controlled comparison of five model families showing that deep-learning models offer no advantage over logistic regression or gradient boosting on these data.
- A quantified answer to the cardiovascular question: in this cohort and setting, cardiovascular and medical history adds about 0.001 AUROC once current cognition is known.
- Results reported separately for cognitively normal participants and participants with MCI, and a measurement of how strongly the prediction depends on current cognitive status.
- A PET image pipeline validated against the PET core's own measurements.
- A research prototype with input plausibility checks and an evidence assistant restricted to published sources.
- A set of independent verification checks that found and corrected two sources of information leakage.

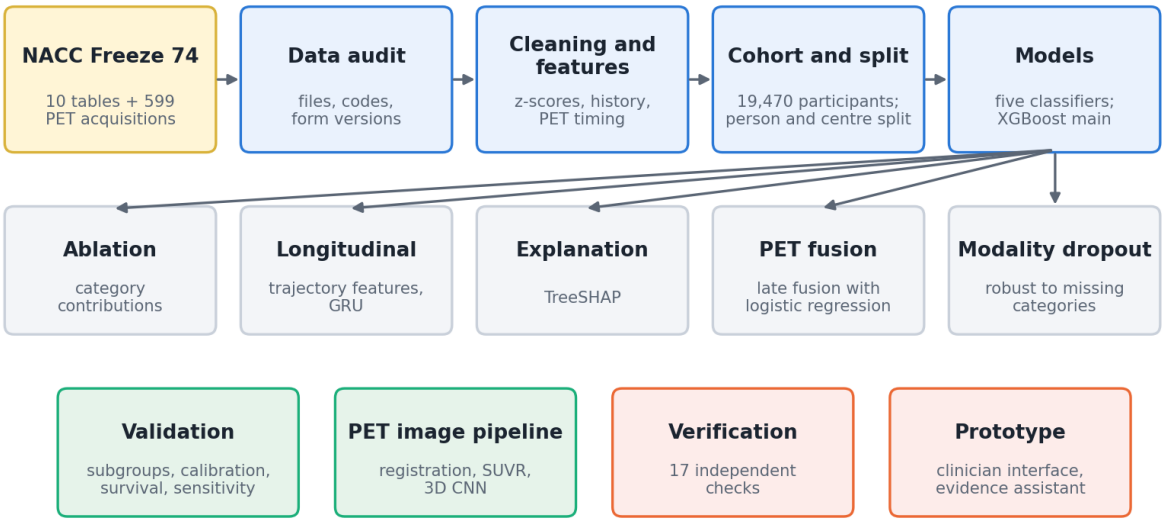


Figure 1.1: Overview of the work presented in this thesis.

1.7 Organisation of the Thesis

Chapter 2 reviews related work. Chapter 3 describes the data. Chapter 4 presents the prediction task, cohort, features, models and evaluation. Chapter 5 describes the PET image pipeline. Chapter 6 reports the results. Chapter 7 describes the prototype. Chapter 8 describes the verification of the pipeline. Chapter 9 discusses the findings and their limitations, and Chapter 10 concludes. The appendices list the features and the reproduction steps.

Chapter 2

Literature Review

2.1 Dementia, Mild Cognitive Impairment and Their Definition

Clinical diagnosis of dementia rests on a decline in cognition that interferes with independence in daily activities, whereas MCI denotes measurable impairment with preserved independence. Alzheimer's disease, the most common cause, has increasingly been defined biologically rather than by its clinical presentation. The Alzheimer's Association workgroup updated the 2018 research framework in 2024, defining the disease by biomarkers of its characteristic pathology, amyloid plaques and tau tangles, measured in fluid or by PET, and staging it separately from the clinical syndrome [8]. This distinction matters for prediction studies: an outcome defined by clinical diagnosis, as in this thesis, measures progression of the syndrome, not of the underlying pathology.

2.2 Modifiable and Cardiovascular Risk Factors

The 2020 report of the Lancet Commission on dementia prevention described a life-course model with twelve potentially modifiable risk factors, including hypertension, diabetes, obesity, smoking and physical inactivity, which together account for around 40% of dementias worldwide [1]. Hypertension, particularly in midlife, has been related to a higher risk of cognitive decline and dementia, although the association is complex and differs with age at measurement [2]. Hypertension is the main risk factor for cerebral small vessel disease, a leading vascular contributor to cognitive impairment [3]. A recent meta-analysis found that higher cumulative systolic blood pressure was associated with incident dementia (odds ratio 1.21, 95% CI 1.00–1.45), while associations with diastolic pressure were inconsistent [9]. For diabetes, a meta-analysis of 15 studies reported a pooled relative risk of dementia of 1.59 (95% CI 1.40–1.80) [4], and among people with MCI, diabetes or the metabolic syndrome was associated with faster progression to dementia (pooled odds ratio 1.67, 95% CI 1.27–2.19) [10]. In people without dementia, a higher Framingham car-

diovascular risk score has been associated with more small vessel disease and with lower cerebrospinal fluid amyloid-β, with small vessel disease mediating the association [11].

These studies establish associations, mostly with exposure measured in midlife or accumulated over many years. They do not by themselves show how much cardiovascular information adds to a prediction model that already contains measurements of current cognition, which is the question addressed in this thesis.

2.3 Genetic Risk

The $\varepsilon 4$ allele of the APOE gene is the major genetic risk factor for late-onset Alzheimer's disease; APOE is the main cholesterol carrier in the brain and is involved in lipid transport and synaptic function [5]. In cohort data the genotype is not always available, and whether a participant was genotyped is not random, which has to be handled explicitly when building features.

2.4 Amyloid and Tau PET

PET with amyloid tracers measures cortical amyloid deposition, tau tracers measure neurofibrillary tangle pathology, and FDG measures glucose metabolism as an index of neuronal activity [6]. Tau PET is technically more demanding than amyloid PET, with tracer-specific binding properties and off-target signal [7]. Because amyloid SUVR values depend on the tracer and the processing pipeline, the Centiloid scale was introduced to express amyloid burden on a common scale anchored at young controls (0) and typical Alzheimer's dementia (100) [12]. These properties motivate the use of Centiloids for amyloid and within-tracer standardisation for tau in this thesis.

2.5 Machine Learning for Tabular Clinical Data

Clinical cohort data are tabular: a mixture of continuous, ordinal and categorical variables with substantial missingness. Logistic regression remains the reference method in clinical prediction because it is transparent and often well calibrated. Ensembles of decision trees, such as random forests [13] and gradient-boosted trees as implemented in XGBoost [14], capture non-linear effects and interactions and handle missing values naturally. Deep-learning models for tables have been proposed repeatedly; in a systematic comparison, Gorishniy et al. introduced the FT-Transformer, which embeds each feature as a token and applies self-attention, and found it to be a strong deep baseline while also observing that no deep model was universally superior to gradient-boosted trees [15]. For sequences of visits, recurrent networks such as the gated recurrent unit [16] can read a participant's history directly.

Evaluation of prediction models requires more than discrimination. Calibration, clinical usefulness assessed by decision curve analysis [17], and robustness to censoring, handled with survival methods such as the Kaplan–Meier estimator [18] and the Cox model [19], are all needed before a model can be interpreted.

2.6 Explainability

Shapley additive explanations (SHAP) attribute a prediction to the input features using Shapley values from cooperative game theory, with the property that the contributions sum to the prediction [20]. For tree ensembles, exact values can be computed efficiently with TreeSHAP [21]. For convolutional networks, gradient-weighted class activation mapping (Grad-CAM) highlights image regions that influence the output [22]. In both cases the explanation describes what the model uses, not what causes the outcome.

2.7 Retrieval-Augmented Generation

Language models can produce fluent but unsupported statements, which is a particular risk in medicine. Retrieval-augmented generation conditions the generated text on passages retrieved from a defined collection [23]. Retrieval can use sentence embeddings [24], keyword ranking functions such as BM25 [25], or a combination of rankings, for which reciprocal rank fusion is a simple and effective method [26].

2.8 Related Prediction Studies

Several studies have combined cardiovascular, clinical and genetic information to predict dementia. In the NACC database, Alaka et al. added resting heart rate to the CAIDE cardiovascular dementia risk score using a random forest and reported improved discrimination in most racial and ethnic groups (mean AUC 0.80–0.91), but not in American Indian participants [27]. A systematic review of 22 studies found that 14 of 15 cardiovascular risk models were associated with dementia or cognitive decline, but that comparisons between models were hampered by differences in cohorts, tests and designs [28]. In the Rush Memory and Aging Project, a higher Framingham cardiovascular risk score was associated with faster decline in global cognition and memory and with smaller hippocampal volume [29].

Among machine-learning studies, Stern and Linial predicted Alzheimer's disease in the UK Biobank from diagnosis codes, APOE genotype and modifiable risk factors, with CatBoost performing best (ROC-AUC 0.773) [30]. Aschwanden et al. ranked 52 predictors of cognitive impairment and dementia in the Health and Retirement Study with random survival forests and Cox models, and found that cardiovascular factors and polygenic scores were less important than expected when evaluated together with sociodemographic,

psychological and health variables [31]. For diagnosis rather than prediction, Yi et al. classified normal cognition, MCI and Alzheimer's disease with a class-weighted XGBoost model explained by SHAP, reporting AUCs of 0.91 on ADNI and 0.88 on NACC [32], and El-Sappagh et al. combined eleven modalities from ADNI in a two-layer random forest model for diagnosis and progression detection with explanations for each decision [33].

These studies show that cardiovascular information is associated with cognitive outcomes and that tree ensembles with SHAP are effective for these data. Fewer studies measure the added value of cardiovascular information after current cognitive status is accounted for, and few report results separately for cognitively normal participants and participants with MCI.

2.9 Data Source

The National Alzheimer's Coordinating Center maintains the Uniform Data Set, a standardised longitudinal assessment administered at Alzheimer's Disease Research Centers across the United States. Version 3 of the neuropsychological battery, introduced in 2015, replaced several proprietary tests with new ones [34], which creates the harmonisation problem addressed in Chapter 4.

2.10 Research Gap

The epidemiological literature establishes that vascular risk factors, APOE and amyloid and tau pathology are associated with dementia. Prediction studies frequently report high discrimination, but such figures depend strongly on who is included and on whether current cognitive status is among the inputs. Three questions are less often answered together on one large cohort under a strict, leakage-controlled design: how much each source of information adds once current cognition is known, whether deep-learning models improve on classical ones for this kind of data, and how results differ between cognitively normal participants and participants with MCI. This thesis addresses these questions on the NACC database, reports results by baseline diagnosis, and accompanies the model with a validated PET image pipeline, a clinician-facing prototype and independent verification of the pipeline.

Chapter 3

Data

3.1 The National Alzheimer's Coordinating Center

The study uses the Freeze 74 release (June 2026) of the National Alzheimer's Coordinating Center (NACC) database. NACC pools longitudinal data from Alzheimer's Disease Research Centers across the United States. Volunteers are assessed roughly once a year with the Uniform Data Set (UDS), a fixed battery of questionnaires, examinations and neuropsychological tests that every centre administers in the same way [34]. The release used here contains 217,598 visits from 57,038 participants at 46 centres, collected between June 2005 and May 2026. Participants are referred through memory clinics or volunteer directly, so the cohort is a research sample and not a population sample: at the first visit the median age is 71 years, median education is 16 years and 76% of participants identify as White.

Access to the data is governed by the NACC data use agreement. All analysis was performed on a local machine. No participant-level value, identifier or image file name appears in this thesis, in the code repository or in any shared report; every reported number is an aggregate.

3.2 Files received

Table 3.1 lists the ten tables in the release. Only the main clinical table and the four summary PET tables contribute model inputs; the remaining files were audited but do not supply features.

Table 3.1: Files in the NACC release and their role in this study.

File	Rows	Participants	Role
Main clinical table (`ftldlbd`)	217,598	57,038	All clinical features and the outcome
SCAN amyloid PET summary	3,933	3,566	Centiloids and amyloid status
Mixed-protocol amyloid PET summary	2,235	2,073	Same, for older scans
SCAN tau PET	2,653	2,405	Meta-temporal and entorhinal SUVR
Mixed-protocol tau PET	671	616	Same, for older scans
SCAN and mixed-protocol amyloid PET, regional	6,168	—	Not used (tracer-specific SUVR)
SCAN FDG PET	885	639	Not used (about 600 linked participants)
SCAN PET quality-control list	8,759	4,585	Audit of the local image folders
CLARiTI study tracking	857	776	Not used (administrative)

In the clinical table one row is one visit; in the PET tables one row is one scan. The participant identifier is the only key shared by all tables. Because a scan is not tied to a visit number, each scan has to be matched to a visit by date (Section 4.3.4).

In addition, 599 raw PET acquisitions (318 amyloid, 232 tau and 49 FDG scans from 446 participants, about 100 GB of DICOM files) were available locally. They are used in Chapter 5.

3.3 The clinical table

The clinical table has 2,649 columns, but most of them are empty for most visits: modules for frontotemporal and Lewy body disease, autopsy findings, form version 4 items and free text. Only 787 columns contain data in at least 10% of visits. From these, 114 variables were mapped by hand to a meaning, a category, a role and their own missing-value codes, following the NACC Researcher's Data Dictionary. Seventy-six of them are used to build the model inputs.

3.3.1 Missing-value codes

NACC encodes non-responses with numeric codes whose meaning depends on the variable: -4 marks a question that was not part of the visit, codes such as 8, 88 or 888 mark "not applicable", and 9, 99 or 999 mark "unknown". These codes cannot be removed globally. The value 88, for example, is a genuine age in 3,655 visits and a genuine diastolic pressure in 3,090 visits, but means "never smoked" in the years-smoked column (73,669 visits). Cleaning was therefore performed column by column from the variable map.

3.3.2 Changes in the forms

The UDS has been revised three times. The most important change for this study took place with version 3 in 2015, when the neuropsychological battery was replaced: the MMSE, Logical Memory, Digit Span and Boston Naming tests were replaced by the MoCA, Craft Story, Number Span and MINT tests. Only category fluency and the Trail Making tests are common to all versions. The same participant can therefore have different tests at different visits. Health history is also recorded on two forms: the participant-reported history (form A5) and, in version 3, the clinician-assessed conditions (form D2). Telephone visits contain no neuropsychological tests.

3.4 Feature categories

The variables used are grouped into five categories (Table 3.2). The complete list, with source columns and value codes, is given in Appendix A.

Table 3.2: Feature categories, source columns and number of model features.

Category	Main source columns	Features
Demographics	`NACCAGE`, `NACCSEX`, `EDUC`, `NACCNIHR`, `NACCHISP`, `NACCLIVS`	8
Cardiovascular and medical	Twelve conditions from forms A5 and D2, smoking, blood pressure, heart rate, BMI, four medication classes and the medication count	25
Cognitive tests	Twenty neuropsychological tests, the Geriatric Depression Scale and twelve NPI-Q behaviour items	9
Clinical stage	`CDRSUM`, `NACCUDSD`	3
Genetics	`NACCNE4S`, `NACCFAM`	3
PET (second-stage model only)	`CENTILOIDS`, amyloid status, meta-temporal and entorhinal tau SUVR	4

Four groups of variables were deliberately excluded. Autopsy findings are known only after death and would leak future information. The Functional Activities Questionnaire, the level of independence, subjective and informant-reported memory complaints and the

use of Alzheimer's medication are so close to the diagnosis itself that including them would partly reproduce the clinician's decision; they were evaluated separately as a sensitivity analysis. Regional PET values were not used because they are tracer-specific and highly redundant. Form version 4 visits (3% of visits) were excluded because they use new variable names.

3.5 PET data

Amyloid PET values are available for about 5,100 participants who also appear in the clinical table, and tau PET values for 2,724. The amyloid tables provide the Centiloid value, which is comparable across tracers, and a positive or negative status assigned by the PET core. The tau tables provide standardised uptake value ratios (SUVR) in about 160 regions; the meta-temporal composite and the entorhinal cortex were retained as the standard summaries of early tau deposition. Because the two tau tracers (flortaucipir and MK-6240) give different SUVR scales, each tau value was standardised within tracer against amyloid-negative participants from the training set. Only scans that passed quality control were used.

Most PET scans are recent, so few scanned participants have three years of clinical follow-up after the scan. This limits what can be learned from PET in this cohort (Section 6.5).

Chapter 4

Methodology

4.1 Prediction task

The prediction is made at an *index visit*, using only information available at or before that visit. For a participant without dementia at the index visit, the task is to estimate the probability of a clinical diagnosis of dementia within three years:

$$\hat{p} = P\left(\text{dementia diagnosed in } (t_0, t_0 + 3 \text{ years}] \mid \mathbf{x}_{t_0}\right), \tag{4.1}$$

where t_0 is the date of the index visit and $\mathbf{x}_{t_0}$ the feature vector at that visit. The main analysis uses each participant's first visit as the index visit.

4.1.1 Choice of horizon and outcome

Clinical diagnosis (`NACCUDSD`) and the Clinical Dementia Rating are recorded at every visit in every form version, which makes them the most consistent outcomes available. A dementia diagnosis was chosen as the main outcome because it is clinically meaningful and easy to interpret. The three-year horizon balances the number of events against loss to follow-up: Kaplan–Meier estimates [18] show that about 35% of participants with MCI progress to dementia within three years, and that 11% of cognitively normal participants progress to MCI or dementia, while the number of participants with a known status falls steadily with longer horizons and most PET scans are too recent for a five-year horizon.

4.1.2 Censoring

A participant last seen less than three years after the index visit without a dementia diagnosis has an unknown outcome. Such participants were excluded from the binary task rather than counted as negative, which would bias the estimated risk downwards. Because this exclusion could itself introduce bias, a Cox proportional hazards model [19] that uses all eligible participants, including the censored ones, was fitted as a robustness check (Section 6.8). Figure 4.1 illustrates the three possible cases.

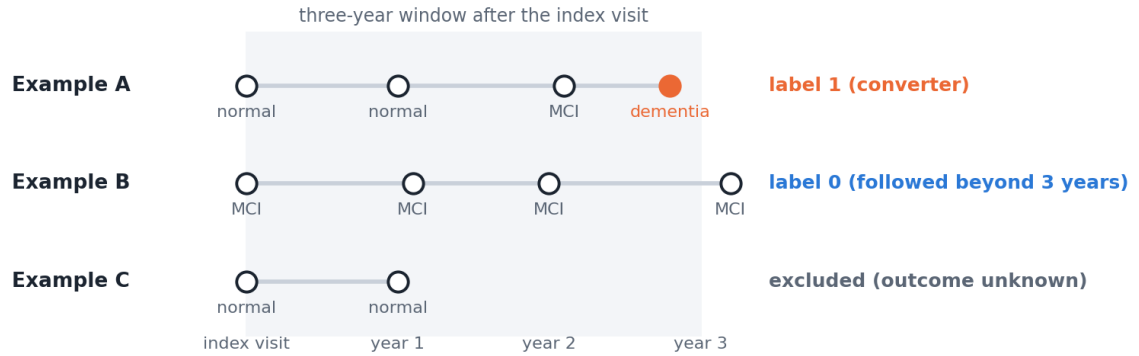


Figure 4.1: Definition of the outcome. Only the index visit supplies features; later visits supply the label.

4.2 Cohort

Table 4.1 shows how the cohort was formed. Visits had to be in person and on form versions 1 to 3, because telephone and version 4 visits lack the neuropsychological tests. Participants with dementia at the index visit, or at any earlier visit, are not eligible.

Table 4.1: Cohort flow.

Step	Participants
All participants	57,038
First visit in person, form version 1 to 3	55,461
and not demented at that visit	38,107
and outcome known at three years	19,470
of whom diagnosed with dementia within three years	2,818 (14.5%)

The "not demented" cohort contains three clinically different groups: 12,377 cognitively normal participants (198 converters, 1.6%), 5,993 participants with MCI (2,485 converters, 41.5%) and 1,100 participants classed as impaired but not MCI (135 converters, 12.3%). Results are reported for the whole cohort and for the normal and MCI groups separately.

4.3 Feature construction

4.3.1 Harmonised cognitive scores

To place tests from different form versions on one scale, each raw score s was converted into a demographically adjusted z-score. For each test, a linear model of the score on age, sex and education was fitted among cognitively normal participants at their first visit, using the training set only:

$$\hat{s} = \beta_0 + \beta_1 \,\text{age} + \beta_2 \,\text{female} + \beta_3 \,\text{education}, \qquad z = \frac{s - \hat{s}}{\sigma_{\text{res}}}, \tag{4.2}$$

where σ_{res} is the residual standard deviation of the fitted model. The sign is reversed for the two timed Trail Making tests, so that a negative z always means worse than expected. Values were clipped to $[-5, 5]$. The z-scores of the tests in one cognitive domain were then averaged to give six domain scores: global screening (MoCA, MMSE), memory, attention, speed and executive function, language, and visuospatial function. A participant receives a domain score from whichever tests their visit contained.

As an example, for the MoCA the fitted model is $\hat{s} = 23.0 - 0.059\,\text{age} + 0.33\,\text{female} + 0.42\,\text{education}$ with $\sigma_{\text{res}} = 2.56$, estimated from 6,264 normal participants. A 72-year-old woman with 16 years of education has an expected score of 25.8, so a score of 21 corresponds to $z = -1.9$ (Figure 4.2).

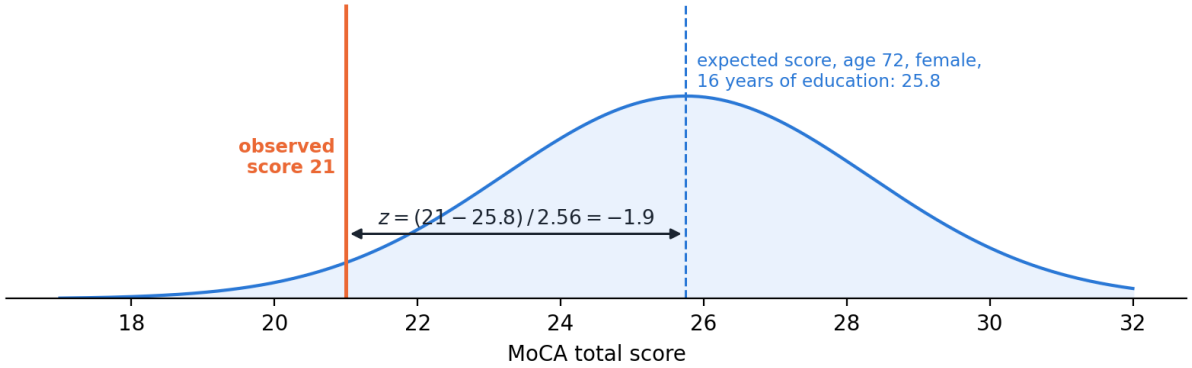


Figure 4.2: Demographically adjusted z-score for the MoCA, using the fitted normative model.

Three further cognitive features were derived: the number of tests the participant could not complete for cognitive or behavioural reasons (NACC code 96 or 996), which is informative about severity and is therefore kept rather than treated as missing; the Geriatric Depression Scale total; and the number of NPI-Q behavioural symptoms present.

4.3.2 Cardiovascular history

Each condition was coded as "ever present" from the participant-reported history (form A5, where both recent and remote mentions count) and, where that form was blank, from the clinician-assessed form D2. Once present, a condition remains present at all later visits. Measured blood pressure, heart rate and BMI use the most recent measurement up to the visit. Pulse pressure and a count of four vascular risk factors (hypertension, hypercholesterolaemia, diabetes and smoking) were derived.

4.3.3 Genetics

The number of APOE $\varepsilon 4$ alleles is used directly. About one participant in five was not genotyped; genotyping depends on centre and consent and is therefore not random, so

these participants receive a separate "not genotyped" indicator instead of an imputed value.

4.3.4 Preventing the use of future information

No feature at a visit may depend on a later visit. Fixed characteristics such as education are carried forward from the visit where they were first recorded, never backwards. A PET scan is used at a visit only if it was acquired on or up to 365 days before that visit. Both rules were introduced after the verification step described in Chapter 8 found violations in an earlier version of the pipeline.

4.4 Data splits

All splits were made at participant level, so that all visits of a person fall in the same partition. Twenty percent of the centres (9 of 46) were held out entirely to measure performance at sites the model has never seen. The remaining participants were divided into training (70%), validation (15%) and test (15%) sets, stratified by diagnosis at the first visit. The random seed was fixed (2026). Table 4.2 gives the resulting sizes for the first-visit cohort.

Table 4.2: Partition sizes for the main cohort.

Partition	Participants	Converters
Training	10,879	1,599
Validation	2,241	342
Test	2,286	332
Held-out centres	4,064	545

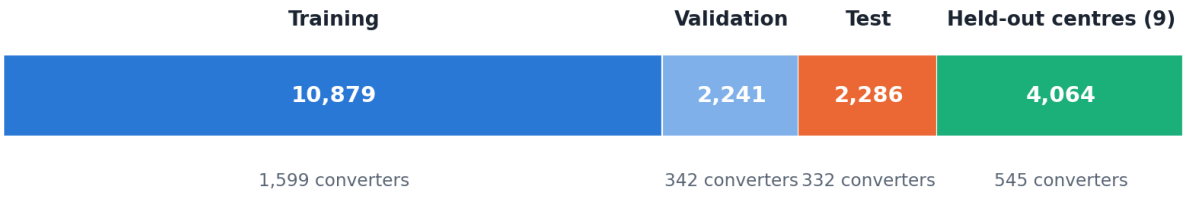


Figure 4.3: Participant-level partitions of the main cohort.

Every quantity estimated from data (normative models for z-scores, tau standardisation, imputation medians, scaling) was fitted on the training set and applied unchanged to the other sets. The validation set was used for early stopping and model selection; the test set and the held-out centres were used only for evaluation.

4.5 Models

Five classifiers of increasing complexity were compared on the same 48 features (Table 4.3). Logistic regression is the standard baseline in clinical prediction. Random forests [13] and gradient-boosted trees (XGBoost [14]) are strong learners for tabular data that handle non-linear effects and interactions. A multilayer perceptron and the FT-Transformer [15] represent deep learning; the latter tokenises each feature and applies self-attention, and is among the strongest deep architectures reported for tabular data. Except for logistic regression, every model was trained with three random seeds and the predictions were averaged.

Table 4.3: Models and settings.

Model	Settings
Logistic regression	Median imputation with missing-value indicators, standardisation; regularisation strength chosen from five values on the validation set
Random forest	500 trees, minimum 10 samples per leaf, $\sqrt{p}$ features per split
XGBoost	Up to 3,000 trees, learning rate 0.03, depth chosen from {3, 4, 6} on validation AUROC, subsample and column sample 0.8, early stopping after 100 rounds without improvement; missing values handled natively
Multilayer perceptron	Two hidden layers (128 and 64 units), dropout 0.2, AdamW, early stopping on validation AUROC
FT-Transformer	Feature tokens of dimension 64, three Transformer layers with eight heads, a learned token for each missing feature, early stopping on validation AUROC

XGBoost was used as the main model for all further experiments because it performed as well as the others, handles missing values without imputation and allows exact Shapley value explanations.

4.6 Contribution of each feature category

The contribution of a category was measured by ablation: the model was trained with all categories and again without the category of interest, and the paired difference in AUROC on the same participants was computed with bootstrap confidence intervals. Each category was also evaluated alone and in cumulative order.

4.7 Longitudinal models

To test whether earlier visits add information, the index visit was moved to a later "landmark" visit preceded by at least two earlier visits. Three models were compared on the same landmarks: XGBoost on the landmark visit only; XGBoost with slope and change features of the cognitive scores and CDR-SB over the earlier visits; and a gated recurrent

unit (GRU) [16] with 48 hidden units that reads up to six visits, with the time of each visit as an input because visit intervals are irregular.

4.8 PET late fusion

Because few scanned participants have three years of follow-up, a large model cannot be fitted on PET participants alone. A late-fusion design was used: the clinical model, trained on all participants, produces a risk for each scanned participant; a small, strongly regularised logistic regression then combines the logit of that risk with the PET measures and indicators of which measures are present. Two rules for matching scans to visits were compared: a strict rule (scan on or before the index visit) and a nearest-visit rule (larger, but the scan may follow the visit by a few weeks).

4.9 Modality dropout

In practice not every patient has every assessment. A model trained on complete records may behave poorly when an entire category is missing. To make one model usable with any combination of available information, the training data were augmented with four additional copies in which each optional category (genetics, cardiovascular, cognitive tests, clinical stage) was removed independently with probability 0.35. This model was compared with a standard model and with specialist models retrained for each pattern of missing categories.

4.10 Explainability

Predictions were explained with Shapley additive explanations (SHAP) [20], computed exactly for the tree ensemble with TreeSHAP [21]. For each test participant, SHAP decomposes the predicted log-odds into a sum of feature contributions. Global importance is the mean absolute contribution of a feature over the test set; category importance is the sum over its features. Stability was assessed by refitting the model on eight bootstrap samples of the training set.

4.11 Evaluation

Discrimination was measured by the area under the ROC curve (AUROC) and the area under the precision–recall curve (AUPRC). Calibration was assessed by the Brier score, calibration slope and intercept, and by comparing predicted and observed risk in deciles. Clinical usefulness was assessed by decision curve analysis [17] and by sensitivity, specificity and predictive values at fixed risk thresholds. Uncertainty was quantified by 1,000

bootstrap resamples of the evaluation set; differences between models were computed on the same resamples (paired bootstrap). Performance was examined within subgroups defined by sex, age, race, education, APOE status and baseline diagnosis.

Chapter 5

PET Image Pipeline

This chapter describes a separate experiment on the 599 raw PET acquisitions. Its purpose is to establish whether the images can be processed into reliable quantitative measurements and used by an image model. It does not feed the dementia model, for the reason given in Section 5.6.

5.1 From DICOM series to static volumes

Each acquisition arrives as a dynamic DICOM series of between 188 and 4,134 files: the same volume repeated over several time frames, from about 20 scanner models. For every scan the stored values were converted to activity concentration, slices were grouped into frames and ordered spatially, and the frames whose mid-point fell inside the tracer's standard static acquisition window were averaged, each weighted by its duration. The result is one static three-dimensional volume per scan. The raw files were only read; all outputs were written to a separate directory.

5.2 Spatial registration

Scans differ in matrix size, voxel spacing and head position, so the same voxel does not correspond to the same anatomical location across scans. No structural MRI was available, so a PET-only registration in two passes was used. In the first pass each scan was aligned to the MNI152 T1 template with a rigid transform (rotation and translation), using mutual information as the similarity measure because PET and T1 images have different contrast. The aligned scans of each tracer class were averaged to form a study-specific PET template, and in the second pass each scan was aligned to that template with a full affine transform (12 parameters). Aligning PET to a PET template is more robust than aligning it to an MRI template.

5.3 Amyloid SUVR from the images

For each registered amyloid scan, the standardised uptake value ratio was computed as the mean uptake in the cortical target region divided by the mean uptake in the whole cerebellum, using the standard masks of the Centiloid project [12]. NACC's PET core had computed an SUVR for the same scans with a different, MRI-based pipeline in native space. Agreement between the two therefore provides an independent test of the conversion and registration steps.

Table 5.1: Agreement between image-derived and NACC amyloid SUVR (318 scans).

Measure	Value
Pearson correlation	0.977
Spearman correlation	0.885
Regression slope (NACC on ours)	1.10
Mean difference (ours minus NACC)	−0.043
95% limits of agreement	−0.18 to +0.09

Agreement was high for every tracer (Pearson r between 0.96 and 0.99 for the four tracers with known identity) and every scanner manufacturer. The lower Spearman correlation reflects the many amyloid-negative scans with similar low values, where small differences change the rank order.

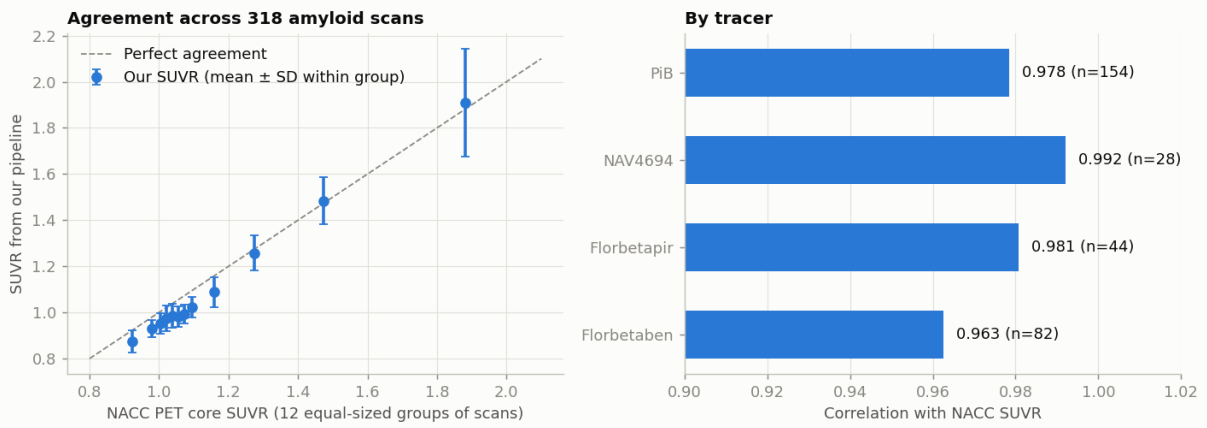


Figure 5.1: Image-derived amyloid SUVR against the value computed by the NACC PET core for the same scans.

5.4 Amyloid status classification

Three classifiers of amyloid status were compared on the 292 amyloid scans with a status label from the PET core (84 positive), using five-fold stratified cross-validation so that every scan was predicted by a model that had not seen it: (i) the cortical SUVR alone; (ii) a logistic regression on mean uptake in an $8 \times 8 \times 8$-voxel grid of blocks; and (iii)

a small 3D convolutional neural network. The network has four stages of two $3 \times 3 \times 3$ convolutions with instance normalisation and ReLU, each stage halving the resolution, followed by global average pooling, dropout of 0.3 and a linear output, with 494,917 weights in total. Volumes were cropped to the head and resampled to $80 \times 96 \times 80$ voxels. Training used AdamW (learning rate 10^{-3}, weight decay 10^{-2}) with a one-cycle schedule, 40 epochs, batches of 8, a class-weighted loss, and augmentation by left–right flips, shifts of up to three voxels and small intensity changes.

Table 5.2: Amyloid status classification (292 scans, 84 positive, cross-validated).

Method	AUROC
Cortical SUVR (one number)	0.982
Regional logistic model	0.971
3D CNN	0.960

The single SUVR value performed best. This is expected: the reference label is itself derived by the PET core from a cortical uptake measure, so a scalar SUVR is close to the definition of the label, and a network trained on fewer than 300 scans has little opportunity to improve on it. The experiment shows that the image pipeline supports a working deep-learning model, not that a CNN is preferable for this task.

5.5 Grad-CAM

Gradient-weighted class activation maps [22] were computed for the network and averaged over 72 scans. The maps were diffuse: about 96% of the volume exceeded a quarter of the maximum activation, and only 1.7% of the most strongly activated voxels lay inside the cortical target region, against 3.8% expected by chance. The maps therefore do not show that the network attends to the cortical regions where amyloid accumulates, and they are not used as evidence of what the model has learned.

5.6 Why images are not used to predict dementia

Of the 446 participants with an image, about 86% were cognitively normal at the time of the scan, follow-up after the scan is about two years, and only five later received a dementia diagnosis. No model can learn a reliable pattern from five events, so dementia prediction from images was not attempted. The quantitative PET information that is available for thousands of participants is used instead, through the late-fusion model described in Chapter 4.

Two further limitations apply. Registration without an MRI is less precise than the MRI-based pipelines used by PET cores. Tau images were converted and registered but

not quantified, because a standard tau target mask comparable to the Centiloid masks was not available.

Chapter 6

Results

All results in this chapter are computed on participants who were not used for training: the internal test set (2,286 participants, 332 converters) and the nine held-out centres (4,064 participants, 545 converters). Intervals are 95% bootstrap intervals. Every number can be traced to a result file in the project repository.

6.1 Comparison of models

Table 6.1: Discrimination and calibration of the five models.

	Test set			Held-out centres	
Model	AUROC (95% CI)	AUPRC	Brier	AUROC	Brier
Logistic regression	0.936 (0.924–0.949)	0.721	0.066	0.945	0.060
Random forest	0.932 (0.919–0.944)	0.704	0.070	0.940	0.064
XGBoost	0.938 (0.925–0.948)	0.713	0.066	0.946	0.059
Multilayer perceptron	0.938 (0.926–0.949)	0.721	0.066	0.946	0.059
FT-Transformer	0.939 (0.927–0.949)	0.720	0.066	0.945	0.061

The five models perform almost identically, and their confidence intervals overlap almost completely (Table 6.1). The deep-learning models did not outperform logistic regression or gradient boosting on these data. For reference, always predicting the training event rate gives a Brier score of 0.124, so all models are substantially better calibrated than a constant prediction.

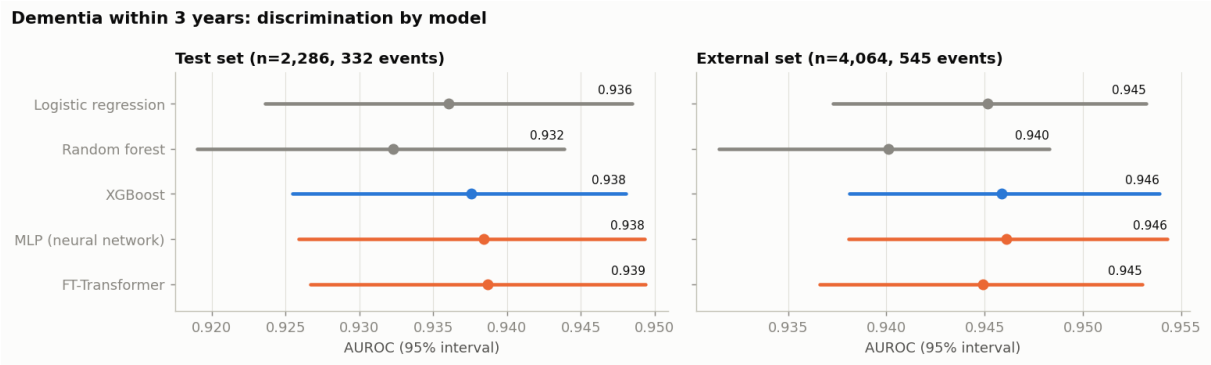


Figure 6.1: AUROC and AUPRC of the five models on the test set and the held-out centres.

6.2 Performance by diagnosis at the index visit

The cohort combines cognitively normal participants, who rarely convert (1.6%), with participants with MCI, who often do (41.5%). Diagnosis alone therefore separates converters from non-converters well: a model given only the diagnosis reaches an AUROC of 0.84. Table 6.2 reports XGBoost separately within each group.

Table 6.2: XGBoost by diagnosis at the index visit (participants / converters).

	Test set		Held-out centres	
Group	*n* / events	AUROC (95% CI)	*n* / events	AUROC (95% CI)
All not demented	2,286 / 332	0.937 (0.925–0.948)	4,064 / 545	0.946 (0.938–0.954)
Cognitively normal	1,411 / 24	0.884 (0.819–0.936)	2,624 / 37	0.876 (0.823–0.918)
MCI	732 / 289	0.835 (0.807–0.862)	1,291 / 496	0.846 (0.824–0.867)

Within MCI the AUROC is about 0.84, which is the more informative estimate of how well the model ranks risk among people who are already impaired. Within the cognitively normal group the AUROC is about 0.88, but it rests on only 24 and 37 events and the intervals are wide; with an event rate of 1.6% the average precision is only 0.15 to 0.19, so most participants flagged as high risk would not convert. Models trained only on normal participants (AUROC 0.876 and 0.881) or only on MCI participants (0.829 and 0.838) gave the same picture, so the group results are not an artefact of mixing the groups during training.

6.3 What drives the prediction

6.3.1 Feature categories

Table 6.3: Contribution of each feature category (XGBoost, test set unless stated).

Category	Alone	Gain when added to all others (test)	Gain (held-out centres)
Cognitive tests	0.896	+0.022 (+0.015 to +0.029)	+0.019 (+0.013 to +0.02
Clinical stage (CDR-SB, diagnosis)	0.897	+0.015 (+0.007 to +0.023)	+0.021 (+0.015 to +0.02
Demographics	0.671	+0.010 (+0.006 to +0.015)	+0.009 (+0.006 to +0.01
Genetics (APOE)	0.645	+0.003 (+0.002 to +0.004)	+0.001 (−0.000 to +0.00
Cardiovascular and medical	0.626	+0.001 (−0.000 to +0.003)	+0.001 (+0.000 to +0.00

Cognitive tests and clinical stage each reach an AUROC near 0.90 on their own and make the largest unique contributions (Table 6.3, Figure 6.2). Because these two categories overlap, removing either one costs little; removing both together, so that the model has no information about current cognition, lowers the AUROC from 0.937 to 0.746 on the test set (0.744 on the held-out centres). Within the normal and MCI groups the corresponding values are 0.783 and 0.638. The model therefore relies mainly on how impaired a person already is at the index visit. Because all these variables are recorded at the index visit and the outcome occurs later, this is not leakage, but it means the model is best described as estimating short-term progression given the current clinical stage.

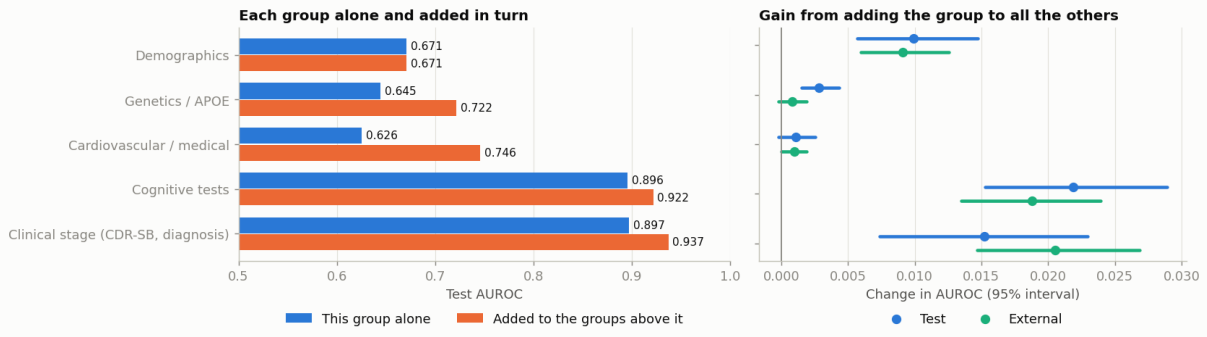


Figure 6.2: Left: AUROC of each category alone and added cumulatively. Right: gain in AUROC from adding each category to all the others.

6.3.2 Cardiovascular information (research question 1)

On its own, the cardiovascular category carries some signal (AUROC 0.626). Added to demographics and APOE only, before any cognitive information, it improved the AUROC by 0.024 on the test set (0.011 to 0.039), but by only 0.004 at the held-out centres (−0.006 to 0.015). Once cognitive information is present its contribution falls to about 0.001, with an interval that includes zero on the test set. Within the cognitively normal group the gain was 0.008 on the test set and 0.003 at the held-out centres, with intervals spanning

zero. In this cohort and prediction setting, therefore, cardiovascular and medical history added little incremental predictive value beyond current cognitive status. This finding does not imply that vascular health is unimportant for the brain: the participants enter at a median age of 71, mid-life exposure is not observed, and vascular damage that has already occurred is likely to be reflected in the cognitive scores.

6.3.3 Shapley explanations

Table 6.4: Share of mean absolute SHAP value by category (test set, 2,286 participants).

Category	Share
Clinical stage (CDR-SB, diagnosis)	40.8%
Cognitive tests	35.0%
Demographics	15.5%
Cardiovascular and medical	5.1%
Genetics	3.5%

The most important individual features were CDR-SB, MCI status, the memory score, age and the executive score. The ranking was stable across eight bootstrap refits (Spearman correlation with the main model 0.89 to 0.97; 8 to 10 of the top ten features shared). Some associations are not causal: lower BMI was associated with higher predicted risk, which is consistent with weight loss in the years before a dementia diagnosis rather than a protective effect of higher weight.

6.4 Longitudinal information

On landmark visits preceded by at least two earlier visits (test: 1,205 participants, 166 events; held-out centres: 2,254 participants, 326 events), adding the visit history produced only small changes (Table 6.5).

Table 6.5: Effect of earlier visits (paired gain over the landmark-only model).

Model	Test AUROC (gain)	Held-out AUROC (gain)
XGBoost, landmark visit only	0.919	0.953
XGBoost with trajectory features	0.922 (+0.003; −0.006 to +0.011)	0.956 (+0.003; −0.001 to +0.007)
GRU over the visit sequence	0.920 (+0.000; −0.011 to +0.010)	0.958 (+0.005; +0.001 to +0.008)

When current scores are known, the path that led to them adds little for a three-year prediction. This does not show that longitudinal modelling is of no value; a longer horizon, an earlier disease stage or more frequent measurement could give a different result.

6.5 PET measures (research question 2)

On its own, amyloid status is strongly associated with progression: among participants with MCI in the held-out sets, about 34% of amyloid-positive participants received a dementia diagnosis within two years, compared with about 7% of amyloid-negative participants (36% and 9% under the strict scan-matching rule). Added to the clinical model, however, the PET measures did not produce a clear improvement under either matching rule. For example, with strict matching and a three-year horizon, the AUROC at the held-out centres was 0.939 without and 0.946 with amyloid (191 participants, 27 events; paired gain −0.007 to +0.024). The final late-fusion model scored 0.964 against 0.944 for the clinical model on 243 held-out scanned participants with 33 events, a gain of 0.020 whose interval (−0.009 to +0.066) includes zero. A modest benefit cannot be excluded, but it could not be demonstrated with the follow-up currently available.

6.6 Validation

6.6.1 Subgroups

Table 6.6: XGBoost AUROC by subgroup.

Subgroup	Test (n / events)	Held-out centres (n / events)
Female	0.950 (1,338 / 162)	0.947 (2,507 / 272)
Male	0.919 (948 / 170)	0.941 (1,557 / 273)
Age under 65	0.976 (467 / 46)	0.958 (1,067 / 56)
Age 65–74	0.952 (963 / 99)	0.956 (1,463 / 191)
Age 75–84	0.904 (732 / 157)	0.937 (1,044 / 191)
Age 85 and over	0.852 (124 / 30)	0.888 (490 / 107)
White	0.933 (1,813 / 273)	0.943 (3,347 / 499)
Black	0.948 (304 / 31)	0.951 (471 / 26)
Education 12 years or less	0.926 (400 / 92)	0.920 (744 / 137)
Education over 16 years	0.951 (937 / 112)	0.960 (1,551 / 188)
APOE ε4 carrier	0.915 (762 / 150)	0.933 (1,402 / 277)

Discrimination was similar across sex, race and education. It was lower in the oldest participants and in APOE ε4 carriers, both groups with a higher baseline risk.

6.6.2 Calibration and clinical usefulness

On the test set the mean predicted risk (14.5%) equalled the observed rate (14.5%), with a calibration slope of 1.01 and intercept of 0.009. At the held-out centres the model over-predicted slightly (15.0% predicted, 13.4% observed; slope 1.10). Using a risk threshold of 20%, the model flagged 22% of test participants with a sensitivity of 83%, specificity of 88%, positive predictive value of 55% and negative predictive value of 97% (held-out

centres: 90%, 87%, 52% and 98%). Decision curve analysis showed a higher net benefit than treating everyone or no one across all thresholds examined, from 2% to 60%, on both evaluation sets.

6.7 Missing information

Table 6.7: Brier score when categories are missing (test set; lower is better).

Available information	Standard model	Modality dropout	Specialist model
Everything	0.066	0.067	0.066
No cognitive tests	0.082	0.077	0.080
No clinical stage	0.185	0.076	0.074
No cognitive information at all	0.441	0.115	0.111
Demographics only	0.444	0.120	0.118

A standard model given missing inputs at prediction time produced badly calibrated risks when cognitive information was absent, worse than predicting the base rate (Brier 0.124). The model trained with modality dropout stayed close to a specialist model retrained for each situation (Table 6.7), with almost no loss when all information is present. This model is used in the clinician interface.

6.8 Sensitivity analyses

Censored participants. A Cox model fitted on all eligible participants, including those followed for less than three years, reached a concordance index of 0.891 on the test set and 0.903 at the held-out centres. Removing the cardiovascular category changed the concordance by 0.001 (−0.001 to 0.003) and 0.000 (−0.001 to 0.002). Hazard ratios per standard deviation were 1.37 for CDR-SB, 0.76 for the memory score, 1.40 for age and 1.20 for each APOE $\varepsilon 4$ allele, while hypertension (0.995, 0.953–1.038), diabetes and stroke had hazard ratios close to one.

Reversed diagnoses. A clinical diagnosis can be revised. Of the 2,818 converters, 217 (7.7%) were recorded as not demented at a later visit. Excluding them, XGBoost reached an AUROC of 0.947 on the test set and 0.950 at the held-out centres (MCI: 0.839 and 0.851). The conclusions do not depend on reversed diagnoses.

Variables close to the diagnosis. Adding the daily-function questionnaire, independence, memory complaints and Alzheimer's medication increased the AUROC by 0.006 on the test set and 0.005 at the held-out centres. These variables were therefore left out of the main model with little loss of accuracy.

Chapter 7

System Design

The models are packaged as a research prototype for a clinician: an interface for entering a patient's information and seeing a risk estimate with its explanation, and an evidence assistant that answers questions from a fixed library of published papers. The prototype is not intended for clinical use and states this on every screen.

7.1 Architecture

Table 7.1: Components of the prototype.

Component	Role
Inference module (`scripts/inference.py`)	Converts raw entries into the 48 model features (for example a MoCA score into the global z-score), runs the modality-dropout model and the PET fusion model, computes SHAP values and checks the plausibility of the entries
API (`app/api.py`, FastAPI)	Serves the form definition, predictions and evidence-assistant answers
Interface (`app/web`, React)	The form and the results
Evidence assistant (`scripts/rag.py`)	Retrieval, guardrails and answer generation

7.2 Input form

The form is generated from the list of features the model was trained on, so it asks for exactly those inputs and nothing else. It has one section per category: demographics (required), cardiovascular history and measurements, the clinician's assessment (cognitive status and CDR-SB), test scores, genetics, and PET. The clinician marks each optional section as available or not. Empty fields are passed to the model as missing values; no default values are substituted. For each test the form shows the expected score of a healthy person of the same age, sex and education.

7.3 Output

The interface reports the estimated three-year risk next to the observed three-year conversion rate of training participants in the same age group and with the same cognitive status, which gives the clinician an empirical reference point. It lists which categories were used and which were missing, and shows the SHAP contributions of each category and of the most influential individual items. When no cognitive information is available, it states that the estimate is little more than an age-based average.

7.4 Plausibility checks

Contradictory entries can produce estimates that look precise but are meaningless. For example, a MoCA score of 0 entered together with a status of "normal" and a CDR-SB of 0 yields a low risk, because the model gives most weight to the clinician's rating. The interface therefore checks the entries against the ranges observed in training data before showing a result. Entries that contradict each other (test scores below the first percentile of cognitively normal participants combined with a normal status; a CDR-SB above the 99th percentile for the stated status; systolic pressure not above diastolic; implausible smoking duration) are flagged and the estimate is marked as unreliable. Unusual but possible entries (age outside the 1st to 99th percentile of the training cohort, very low education, discordant PET values) generate a note.

7.5 Behaviour checks

Twenty-eight automated checks test the deployed model through the same code path as the interface. They confirm that the form covers every trained feature and no other; that typical cognitively normal, impaired and MCI cases receive plausible, correctly ordered estimates (0.6%, 4.0% and 60.8%); that a worse CDR-SB, a worse cognitive status, lower test scores, more APOE $\varepsilon 4$ alleles or more behavioural symptoms never lower the estimate; that contradictory entries are flagged; and that missing categories are reported. Twenty-seven checks pass. The remaining check compares predicted and observed risk within age and diagnosis groups on held-out data: ten of eleven groups agree within seven percentage points, but for participants with MCI under the age of 65 the model predicts 35% against an observed 27%. This over-prediction is reported as a known limitation.

7.6 Evidence assistant

The evidence assistant is a retrieval-augmented generation system [23]. Its library consists of eleven open-access papers on dementia risk factors, vascular contributions, APOE, amyloid and tau PET and diagnostic criteria, divided into 447 passages. For a question,

two rankings are computed, one by semantic similarity of sentence embeddings [24] and one by BM25 keyword relevance [25], and merged by reciprocal rank fusion [26]. If the best semantic similarity is below a fixed threshold (0.35), the assistant reports that the library contains no evidence on the question. Otherwise a small instruction-tuned language model running locally writes an answer from the retrieved passages with numbered citations; if the answer does not cite the passages correctly, the assistant quotes them directly instead. Questions asking for advice about a particular person are answered with a notice that the system cannot give individual medical advice, followed by general published information only. No data leave the machine.

Table 7.2: Evaluation of the evidence assistant (52 questions).

Measure	Result
Correct source ranked first (hybrid retrieval)	32 of 32
Correct source ranked first (embeddings only / BM25 only)	28 of 32 / 31 of 32
Personal-advice questions flagged	8 of 8
General questions wrongly flagged as advice	0 of 6
Out-of-scope questions declined	6 of 6
Answers written with valid citations by the language model	22% (otherwise quoted)

These results should be read with care. The library is small and the test questions were written by the project team, so retrieval is an easy test. The local language model is small and produced a properly cited answer for only about one question in five; direct quotation is the fallback and is the safer behaviour for medical content.

Chapter 8

Verification and Reproducibility

Clinical prediction studies are vulnerable to subtle errors that inflate performance: the same person in training and test data, information from the future in the features, mislabelled outcomes, or reported numbers that do not match the saved predictions. A separate verification program (`scripts/verify_results.py`) was written to test for these errors independently of the main pipeline. Each check recomputes a quantity in a second, simpler way or tests a property that must hold if the pipeline is correct.

8.1 Checks

Table 8.1: Independent verification checks and their results.

Group	Check	Result
A	Raw files unchanged since the first audit; all 599 image folders present	Pass
B	No participant in more than one partition; no held-out centre in the training data; saved split matches the modelling table	0 participants and 0 centres shared
C	Features at the first and third visit are identical when all later visits are deleted before the features are built	Identical for 1,500 participants × 67 features at each visit
C	No PET value is newer than the visit at which it is used	0 of 1,481 amyloid and 713 tau values
D	Outcome labels recomputed with an independent implementation	0 disagreements in 11,403 visits
D	No eligible index visit has a dementia diagnosis	0 of 134,943
E	With shuffled outcome labels, a retrained model falls to chance	AUROC 0.508
F	Reported AUROC values recomputed from the saved predictions	Largest difference 0 over 5 models and 2 evaluation sets
F	Cohort sizes in the reports match the modelling table	Match
G	Retraining from scratch with a new seed reproduces the result	Test 0.937, held-out 0.946 (reported 0.938, 0.946)
G	Gain from the cardiovascular category with a fresh seed	+0.0016
H	Cohort flow and PET index counts in the reports match the data	Match

All seventeen checks passed on the final pipeline (re-run on 22 October 2026).

8.2 Errors found and corrected

The checks found two genuine errors in an earlier version of the pipeline, both of which used information that would not have been available at the time of prediction:

1. Missing education and race at an early visit had been filled in from a later visit for about 0.1% of participants. Fixed characteristics are now carried forward only.

2. PET scans had been matched to the nearest visit, which for many participants was a visit a few weeks before the scan. A scan is now used only at a visit on or after its acquisition date.

After both corrections every model was retrained and every result regenerated. The main results changed only in the third decimal place. Earlier outputs were retained for comparison.

8.3 Reproducibility

All random seeds are fixed (2026 for the split, 0 to 2 for model training). Package versions are recorded in `requirements.txt`. The pipeline is run by a fixed sequence of scripts documented in the repository; given the NACC data, it regenerates every table in this thesis. Participant-level data and trained models are not distributed, in accordance with the data use agreement; all shared outputs are aggregates, and each report is checked automatically for participant identifiers before it is written.

Chapter 9

Discussion and Limitations

9.1 Interpretation of the Main Result

The headline AUROC of 0.94 must be read together with the composition of the cohort. Participants with MCI converted 25 times as often as cognitively normal participants, and the diagnosis alone reached an AUROC of 0.84. Much of the overall discrimination therefore comes from separating these groups, which a clinician already does. The more informative figures are the within-group results: about 0.84 among participants with MCI, and about 0.88 among cognitively normal participants, the latter based on few events. Removing all information about current cognition lowered the AUROC from 0.94 to 0.75. The model is therefore best described as estimating the short-term risk of progression given the current clinical stage, not as detecting future dementia in people without any signs of impairment.

Including CDR-SB and diagnosis as inputs is not leakage in the technical sense: both are measured at the index visit, and the outcome is a diagnosis made at a later visit. It does mean that the clinical label used as outcome and some of the inputs come from the same assessment process. Results without these inputs are reported for this reason.

9.2 The Cardiovascular Question

The central hypothesis of the original project was that cardiovascular information would improve prediction. In this cohort it added about 0.001 AUROC once current cognition was known, the survival analysis gave the same result, and the hazard ratios for hypertension, diabetes and stroke were close to one after adjustment. This is consistent with, rather than contrary to, the epidemiological literature. Most of the evidence concerns midlife exposure or exposure accumulated over decades [2, 9], whereas NACC participants enter at a median age of 71 and their history is recorded largely as yes/no answers with single blood-pressure readings. Damage that vascular disease has already caused is likely to be reflected in the cognitive scores, which the model sees directly. The find-

ing should therefore be stated as limited incremental predictive value in this cohort and setting, not as evidence that vascular health is unimportant for the brain.

9.3 Comparison with Related Studies

The finding that cardiovascular information adds little once other information is known agrees with Aschwanden et al., who found cardiovascular factors less important than expected when evaluated jointly with other predictors [31]. It does not contradict the association studies [28, 29], which ask whether cardiovascular risk is related to later cognition rather than whether it improves a model that already measures current cognition. Our AUROC is higher than that of risk models without cognitive inputs, such as the UK Biobank model of Stern and Linial (0.773) [30], because our model includes current cognition and clinical stage; without them our model scored 0.75, in the same range. Like Yi et al. [32], we found CDR-SB to be the most important feature, although their task was diagnosis rather than prediction of future conversion.

9.4 Model Complexity

Logistic regression, random forest, XGBoost, a multilayer perceptron and an FT-Transformer performed within 0.007 AUROC of one another, with overlapping intervals. This agrees with comparisons on other tabular data [15]. With a few dozen informative features, most of them strongly related to the outcome, there is little structure left for a more flexible model to exploit. The deep-learning idea that did prove useful was modality dropout, which addresses a practical problem (missing assessments) rather than raising discrimination.

9.5 PET

Amyloid status was strongly associated with progression in participants with MCI, in line with the biological definition of the disease [8], but no clear gain was shown when PET was added to the clinical model. The likely reason is limited follow-up: most scans are recent, and only a few dozen scanned participants with a known three-year outcome converted. The PET image experiment showed that the raw images can be processed into measurements that agree closely with the PET core's, but with five converters among the participants with images, image-based prediction of dementia was not attempted.

9.6 Limitations

1. **Cohort.** NACC is a research cohort recruited through memory centres and volunteers; participants are mostly White and highly educated. Absolute risks and performance may differ in other populations.

2. **Validation.** The held-out centres belong to the same study, use the same forms and were collected in the same period. This is site-level validation, not external validation on an independent cohort, and no temporal validation was performed.

3. **Censoring.** About half of the eligible participants were excluded because their three-year outcome was unknown. Participants with more severe impairment return less often, so the analysed cohort is somewhat healthier than all enrolled participants. The survival analysis that includes censored participants reached the same conclusions.

4. **Outcome.** The outcome is a clinical diagnosis, which can be revised; 7.7% of converters were later recorded as not demented. Excluding them did not change the results.

5. **Cognitively normal participants.** Only 24 test and 37 held-out converters were cognitively normal at the index visit, so performance in this group is uncertain.

6. **Calibration.** Risk was slightly over-predicted at the held-out centres and by about eight percentage points for participants with MCI younger than 65.

7. **Demographic inputs.** Race, ethnicity and living situation improved calibration across racial groups, most likely because of differences in recruitment, but they change individual estimates noticeably. Whether such inputs belong in a clinical tool is a question of policy as well as accuracy.

8. **Test reuse.** The same test set was used for several experiments. The held-out centres provide a second evaluation set, and the results agree.

9. **PET images.** Registration was performed without MRI, tau images were not quantified, and Grad-CAM maps were not informative.

10. **Evidence assistant.** The library contains eleven papers and the test questions were written by the team.

9.7 Ethical Considerations

The data were used under the NACC data use agreement. No participant-level information leaves the analysis machine, and every shared output is checked automatically for identifiers. The prototype is labelled as a research tool, does not diagnose, and does not

recommend treatment: the data record what happened to participants and cannot show what would have happened under a different treatment. The evidence assistant answers only from published sources and declines to give advice about an individual.

Chapter 10

Conclusion and Scope for Further Research

10.1 Conclusion

This thesis set out to estimate, for people without dementia at a clinic visit, the probability of a dementia diagnosis within three years, and to establish which kinds of information contribute to that estimate. Using 19,470 participants from the NACC database under a person-level split with nine held-out centres, the answers to the six research questions are as follows.

RQ1. Three-year conversion to dementia can be predicted with an AUROC of 0.94 in the combined cohort and 0.95 at held-out centres. Within participants with MCI the AUROC is about 0.84, and within cognitively normal participants about 0.88, with wide uncertainty.

RQ2. Cardiovascular and medical history adds about 0.001 AUROC once current cognition is known; it has limited incremental value in this cohort and setting.

RQ3. Amyloid and tau PET measures did not produce a demonstrable gain over the clinical model with the follow-up currently available.

RQ4. Deep-learning models (a multilayer perceptron and an FT-Transformer) performed no better than logistic regression or gradient-boosted trees.

RQ5. Earlier visits added between 0.000 and 0.005 AUROC beyond the current visit.

RQ6. A model trained with modality dropout remained close to specialist models when whole categories of information were missing, whereas a standard model became badly calibrated.

Most of the predictive information lies in the current cognitive and clinical state. The results are robust to the exclusion of participants with unknown outcomes, to reversed

diagnoses and to retraining with different seeds, and the pipeline passed seventeen independent verification checks after two sources of information leakage had been found and corrected. Alongside the models, the work produced a PET image pipeline whose amyloid measurements agree with the PET core's (correlation 0.98), and a research prototype that checks the plausibility of its inputs and supports its output with an evidence assistant restricted to published sources.

10.2 Scope for Further Research

10.2.1 External and Temporal Validation

The model should be evaluated on an independent cohort with different recruitment and on participants enrolled after the training period.

10.2.2 Cognitively Normal Participants and Longer Horizons

A model built specifically for cognitively normal participants, with a five- or ten-year horizon, would have more events and more room for cardiovascular and PET information to contribute.

10.2.3 Richer Cardiovascular Exposure

Repeated blood-pressure measurements over many visits, treatment history and imaging markers of small vessel disease could capture vascular burden better than yes/no history.

10.2.4 PET with Longer Follow-up

As more scanned participants reach three or more years of follow-up, the added value of amyloid and tau PET can be tested with adequate power, and MRI-based processing of the images can improve quantification.

10.2.5 Evidence Assistant

A larger, curated library, an independently written evaluation set and a stronger local language model would make the assistant more useful.

10.3 Closing Observation

A carefully measured negative result is still a result. In a large, well-characterised cohort, the information that best predicts dementia in the next three years is how a person is

functioning today; the factors that shaped the brain over decades leave their trace in that measurement rather than adding to it.

Appendix A

Feature List

Table A.1: The 48 model features and their source columns in the NACC clinical table.

Category	Feature	Source and construction
Demographics	age	NACCAGE
	female	NACCSEX = 2
	education (years)	EDUC, carried forward
	race (White, Black, Asian)	NACCNIHR, three indicators
	Hispanic	NACCHISP
	lives alone	NACCLIVS = 1
Cardiovascular	hypertension, hypercholesterolaemia, diabetes, myocardial infarction, atrial fibrillation, heart failure, angioplasty, bypass, pacemaker, other cardiovascular disease (10)	Form A5 (HYPERTEN, HYPERCHO, DIABETES, CVHATT, CVAFIB, CVCHF, CVANGIO, CVBYPASS, CVPACE, CVOTHR); form D2 where A5 is blank; "ever" carried forward
	stroke, TIA	NACCSTYR, NACCTIYR, with CBSTROKE, CBTIA
	ever smoked, years smoked	TOBAC100, SMOKYRS
	systolic and diastolic BP, heart rate, BMI	BPSYS, BPDIAS, HRATE, NACCBMI, latest measurement
	pulse pressure	systolic minus diastolic
	vascular risk count	hypertension + hypercholesterolaemia + diabetes + smoking
	medication classes and count (5)	NACCAHTN, NACCLIPL, NACCDBMD, NACCAC, NACCAMD
Cognitive tests	global score	z of MOCATOTS, NACCMMSE
	memory score	z of CRAFTVRS, CRAFTDVR, LOGIMEM, MEMUNITS, UDSBENTD
	attention score	z of DIGFORCT, DIGBACCT, DIGIF, DIGIB
	executive score	z of TRAILA, TRAILB, WAIS
	language score	z of ANIMALS, VEG, UDSVERFC, BOSTON, MINTTOTS
	visuospatial score	z of UDSBENTC
	tests not completed	count of codes 96 / 996
	depression	NACCGDS
	behavioural symptoms	count of twelve NPI-Q items
Clinical stage	CDR-SB	CDRSUM
	MCI, impaired not MCI	NACCUDSD = 3, = 2
Genetics	APOE $\varepsilon 4$ count, not genotyped	NACCNE4S
	family history	NACCFAM

Appendix B

Implementation and Reproducibility Notes

B.1 Software

Python 3.11 with the package versions listed in `requirements.txt`: NumPy 1.26, pandas 2.3, scikit-learn 1.6, XGBoost 3.0, PyTorch 2.8, SHAP 0.48, lifelines 0.30, SimpleITK 2.5, pydicom 3.0, sentence-transformers 6.1, FastAPI 0.116 and React 18 for the interface.

B.2 Pipeline

With the NACC files placed under `data/raw/`, the results are regenerated by running, in order: `audit_data.py`, `build_dataset.py`, `train_tabular.py`, `train_by_baseline_group.py`, `train_longitudinal.py`, `explain.py`, `train_pet.py`, `validate.py`, `train_final.py`, `sensitivity_confirmed.py`, the four PET image scripts, the three evidence-assistant scripts, and finally `verify_results.py` and `check_clinical_behaviour.py`.

B.3 Seeds

Data split: 2026. Model training: 0, 1 and 2, averaged. Image cross-validation: 0.

B.4 Data Access

Participant-level data and trained models cannot be distributed under the NACC data use agreement. All reported results are aggregates and are stored in the `reports` directory of the project repository.

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List of Publications

(Based on the Research Work)

No publications or communicated papers were produced from this thesis work at the time of submission.